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Machine Learning-Based Network Intrusion Detection System **with Explainable AI**

Research Abstract:-

Modern networked organisations face a relentless and ever-evolving stream of malicious traffic — port scans probing for open services, distributed denial-of-service floods, brute-force credential attacks, SQL injection attempts, and lateral movement by adversaries already operating within the network perimeter. Traditional signature-based firewalls and intrusion detection tools fail to generalise beyond known attack patterns, leaving infrastructure critically exposed to novel and polymorphic threats. There is a clear need for an intelligent, adaptive detection layer that can classify diverse attack types in real time while remaining interpretable to security teams.

This project presents a ML based Network Intrusion Detection System (NIDS) trained on the CIC-IDS-2017 dataset, comprising 2.8 million network flows across 14 distinct attack categories including DDoS, PortScan, Botnet, Heartbleed, and multiple Web Attack variants. The system aggregates raw packets into 80-feature flow records — capturing metadata, byte counts, packet lengths, inter-arrival times, and TCP flags — eliminating the need for deep packet inspection and ensuring compatibility with encrypted traffic. A Gradient Boosted Trees classifier XGBoost is trained on these features and delivering sub-millisecond inference per flow, enabling real-time classification at 1 Gbps line rate without any GPU dependency. To ensure transparency, SHAP (SHapley Additive exPlanations) values accompany every alert, providing SOC analysts with clear, feature-level reasoning behind each classification and significantly reducing false positive fatigue. An Isolation Forest module runs in parallel to detect anomalous flows that fall outside the classifier's known training distribution, preventing novel attacks from being silently passed as benign. The complete system is deployed as a FastAPI microservice integrated with a Streamlit monitoring dashboard, targeting classification accuracy exceeding 98% and a false positive rate below 1% on benign traffic.

Github Repository:- <https://github.com/tensorlj/Network-Intrusion-Detection-System>